

Software Checklist

Please be ready with these before Next Saturday 15th Nov'25 class

- ☐ Python Distribution
- ☐ IDE
- ☐ GIT → GIT BASH
- ☐ DOCKER
- ☐ POSTMAN (Optional)
- ☐ WARP (Optional)
- ☐ AWS CLI

Python Distributions(Any One)

- ☐ Anaconda (*Majorly we will use in this course*)
- ☐ Python

Download Python

The official home of the Python Programming Language

 <https://www.python.org/downloads/>



Download Now | Anaconda

Anaconda is the birthplace of Python data science. We are a movement of data scientists, data-driven enterprises, and open source communities.

 <https://www.anaconda.com/download/success>

Windows Env variables PATH:

C:\Users\Admin\anaconda3
C:\Users\Admin\anaconda3\Library\mingw-w64\bin
C:\Users\Admin\anaconda3\Library\usr\bin
C:\Users\Admin\anaconda3\Library\bin
C:\Users\Admin\anaconda3\Scripts

Package Managers(Any One)

- ☐ Conda (*Rarely*)
- ☐ pip (*Majorly we will use in this course*)
- ☐ uv (*Optional*)

Refer for UV

UV- Python Package And Project Manager- Faster Than Pip

An extremely fast Python package and project manager, written in Rust.

<https://docs.astral.sh/uv/>

Highlights

 <https://www.youtube.com/watch?v=JeiCxJP7IK4>



Local IDE or Text Editor(Any One)

- ☐ VSCODE (*Majorly we will use in this course*)
- ☐ CURSOR PAID
- ☐ WINDSURF PAID
- ☐ TRAE
- ☐ PYCHARM Community or Professional
- ☐ KIRO

<https://code.visualstudio.com/>

Cursor - The AI Code Editor

Built to make you extraordinarily productive, Cursor is the best way to code with AI.

 <https://www.cursor.com/>



Windsurf (formerly Codeium) - The most powerful AI Code Editor

Windsurf (formerly Codeium) is the world's most advanced AI coding assistant for developers and enterprises. Windsurf Editor — the first AI-native IDE that keeps developers in flow.

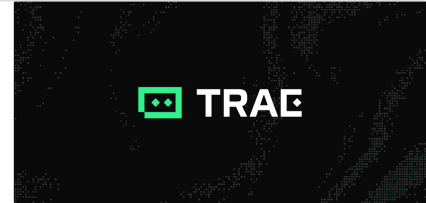
 <https://windsurf.com/>


The Windsurf Editor
[Built to keep you in the flow state]

Trae - Collaborate with Intelligence

Trae IDE integrates seamlessly into your workflow, collaborating with you to maximize performance and efficiency.

 <https://www.trae.ai/>



PyCharm: The only Python IDE you need

Built for web, data, and AI/ML professionals. Supercharged with an AI-enhanced IDE experience.

 <https://www.jetbrains.com/pycharm/>



OPTIONAL : Browser based Online IDE

- ☐ Lightning.ai
- ☐ Replit
- ☐ Firebase Studio

Working with virtual environment

- ☐ venv
- ☐ conda
- ☐ uv

Managing environments — conda 25.5.1.dev2 documentation

With conda, you can create, export, list, remove, and update environments that have different versions of Python and/or packages installed in them. Switching or moving between

 <https://docs.conda.io/projects/conda/en/latest/user-guide/tasks/manage-environments.html>

```
# creates the virtual env in anaconda3 folder
conda create -n venv_ur_name python==3.12 -y
conda activate venv_ur_name
conda deactivate
conda remove -n venv_ur_name --all
```

OR

```
# creates the virtual env in current folder
conda create -p venv_ur_name python==3.12 -y
conda activate
conda deactivate
# Delete the folder venv_ur_name
```

Refer :- <https://gist.github.com/loic-nazarie/b18a908473935243fc23586f35d4bacc>

OPTIONAL VENV

🌐 Python : Virtual Environment Management with `venv`

📌 Prerequisite

Ensure Python is installed:

```
```bash
python --version
```
```

📁 1. Create Virtual Environment

Replace `myenv` with your desired environment name.

✅ Windows

```
```bash
python -m venv myenv
```
```

✅ macOS / Linux

```
```bash
python -m venv myenv
```
```

📁 2. Activate Virtual Environment

 Windows (Command Prompt)

```
```bash
myenv\Scripts\activate
```
```

 Windows (PowerShell)

```
```bash
.\myenv\Scripts\Activate.ps1
```
```

 macOS / Linux

```
```bash
source myenv/bin/activate
```
```

> After activation, your shell prompt will change to show the environment name.

 3. Deactivate Virtual Environment

Works the same on all platforms:

```
```bash
deactivate
```
```

 4. Delete Virtual Environment

Simply remove the folder:

 Windows

```
```bash
rmdir /s /q myenv
```
```

 macOS / Linux

```
```bash
```

```
rm -rf myenv
...
```

```

```

### venv — Creation of virtual environments

Source code: [Lib/venv/](#) The venv module supports creating lightweight “virtual environments”, each with their own independent set of Python packages installed in their site directories. A virtual en...

 <https://docs.python.org/3/library/venv.html>

Python documentation

### venv — Creation of virtual environments



Source code: [Lib/venv/](#) The venv module supports creating lightweight “virtual environments”, each with their own independent set of Python packages installed...  
[docs.python.org/3/](https://docs.python.org/3/)



## UV

uv

uv is an extremely fast Python package and project manager, written in Rust.

 <https://docs.astral.sh/uv/>

### # 🐍 Python : Virtual Environment Management with `uv` (Cross-platform)

[`uv`](<https://github.com/astral-sh/uv>) is a super-fast Python package manager that also supports creating virtual environments. Here's how to manage virtual environments using `uv` on **Windows**, **Linux**, and **macOS**.

```

```

### ## 📦 Step 1: Install `uv`

Make sure you have Python 3.12 installed.

```
```bash
```

```
# Using pipx (recommended)
```

```
pipx install uv
```

```
# Or via pip
```

```
pip install uv
```

```
# Create environment
```

```
# Syntax
```

```
uv venv <env_name>
```

```
# Example
```

```
uv venv myenv
```

```
# Activate
```

```
myenv\Scripts\activate → Windows
```

```
source myenv/bin/activate → Mac/Linux
```

```
# Deactivate
```

```
deactivate
```

```
...
```

```
-----
```

```
##### OR #####
```

```
-----
```

```
```bash
```

```
uv venv --python 3.11
```

```
source .venv/bin/activate
```

```
Deactivate
```

```
uv pip install -r req.txt
```

```
rm -rf myenv
```

```
Create new uv environment and install
requirements
```

```
uv venv myenv
```

```
source myenv/bin/activate
```

```
uv pip install -r requirements.txt
```

```
uv sync
```

```
uv venv --python 3.11 myenv
```

```
source myenv/bin/activate
```

```
uv pip install -r requirements.txt
```

```
uv add -r requirements.txt
```

```
...
```

```
set Path=C:\Users\aksha\.local\bin;%Path% (cmd)
```

```
$env:Path = "C:\Users\aksha\.local\bin;$env:Path" (powershell)
```

# Version Control System

☐ GIT (*Always we use in the course*)

☐ Github Desktop (*Optional*)

## Git And Github

 <https://www.youtube.com/playlist?list=PLZoTAEaLRMXVOSsBerFZKsdCaA4RYr4RGW>



## Git - Downloads

The entire Pro Git book written  
by Scott Chacon and Ben Straub is available to read online for free.  
Dead tree versions are available on Amazon.com.

 <https://git-scm.com/downloads>

## Commands used in Class

```
Creating conda environment
conda create -n llmops python=3.12 -y

activate conda environment
conda activate llmops

Creating conda env in custom path
For current directory : conda create -p abcd001 python=3.11
For directory of choice : conda create -p c:/abc/usr/abcd002 python=3.11

Activating conda environment created using -p
conda activate c:/abc/usr/abcd002

Creating requirements.txt
pip freeze
pip freeze > requirements.txt
pip list
pip list --format=freeze > req.txt

Installing requirements from requirements file
pip install -r file_name.txt
```



# Extra Tools

## Markdown

### Basic Syntax | Markdown Guide

The Markdown elements outlined in the original design document.

 <https://www.markdownguide.org/basic-syntax/>



## Docker

### Docker Cheatsheet

[https://docs.docker.com/get-started/docker\\_cheatsheet.pdf](https://docs.docker.com/get-started/docker_cheatsheet.pdf)

### Docker Desktop: The #1 Containerization Tool for Developers | Docker

Docker Desktop is collaborative containerization software for developers. Get started and download Docker Desktop today on Mac, Windows, or Linux.

 <https://www.docker.com/products/docker-desktop/>

 **Download Docker Desktop**

## Docker Basics

### The Ultimate Docker Cheat Sheet

Docker - Beginners Intermediate Advanced

<https://dockerlabs.collabnix.com/docker/cheatsheet/>

### Complete Dockers For Data Science Tutorial In One Shot

In this videos we will be seeing how we can implement end to end Data Science Project Implementation

 <https://www.youtube.com/watch?v=8vmKtS8W7IQ>



### System Check for Windows

#### Using Windows Settings

Open Settings: Press Windows + I on your keyboard.

Go to System > About.

Check Device Specifications:

Look for the System Type entry.

If it says "ARM-based processor" or "ARM64-based processor", your PC uses an ARM chip.

If it says "x64-based processor" or "AMD64", your PC uses an AMD (or Intel) processor.

## WSL Setup in Windows

### Windows

Get started with Docker for Windows. This guide covers system requirements, where to download, and instructions on how to install and update.

 <https://docs.docker.com/desktop/setup/install/windows-install/#option-1-install-or-update-wsl-via-the-terminal>



## Markdown Basics

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## Cloud CLI

### AWS CLI

#### Installing or updating to the latest version of the AWS CLI - AWS Command Line Interface

The AWS CLI is an open source tool built using the AWS SDK for Python (Boto) that provides commands for interacting with AWS services. With minimal configuration, you can start using all of the functionality provided by the AWS Management Console from your favorite terminal program.

 <https://docs.aws.amazon.com/cli/latest/userguide/getting-started-install.html>



## GCP CLI

### Install the gcloud CLI | Google Cloud CLI Documentation

This page contains instructions for choosing and maintaining a Google Cloud CLI installation. The Google Cloud CLI includes the gcloud, gsutil and bq command-line tools. For a list of gcloud CLI features, see

 <https://cloud.google.com/sdk/docs/install#windows>



## API Tool

### Postman: The World's Leading API Platform | Sign Up for Free

Accelerate API development with Postman's all-in-one platform. Streamline collaboration and simplify the API lifecycle for faster, better results. Learn more.

 <https://www.postman.com/>



## Warp Terminal

### Warp: The Agentic Development Environment

The fastest way to build with multiple AI agents, from writing code to deploying it. Trusted by over half a million engineers, Warp gives developers speed, privacy, and control to ship faster.

 <https://www.warp.dev/>

